Multi-Algorithm Recommendation System Comparison

E-Commerce Recommendation System Analysis Report

1. Problem Statement

Modern e-commerce platforms rely heavily on personalization to enhance user experience, drive engagement, and boost conversions. A robust recommendation system must address multiple dimensions of customer interaction: predicting potential product ratings (regression), forecasting purchase likelihood (classification), and segmenting users into distinct behavioral cohorts (clustering). This report implements and compares three machine learning algorithms—Ridge Regression, Logistic Regression, and K-Means Clustering—on an e-commerce dataset to deliver actionable business intelligence and optimize recommendation strategies.

2. Dataset Description & Preprocessing

The dataset contains 5,000 transaction records documenting interactions between users and products. Columns include User ID, Product ID, Product Category, Price, Rating, Purchase Status, Number of Views, Cart Status, Time Spent, Previous Purchases, and Interaction Date.

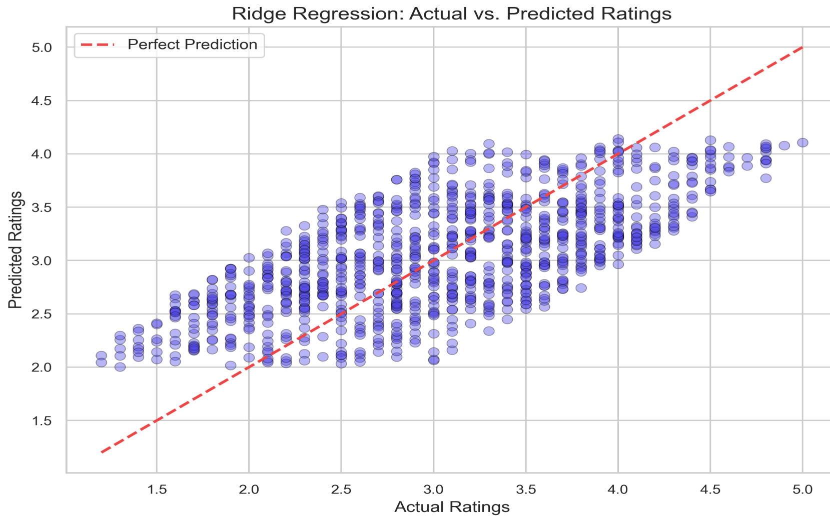
Preprocessing Steps Performed:

- Verified dimensions and checked for duplicate/missing records (none found/handled).
- One-Hot Encoded categorical product categories to construct numerical feature vectors.
- Split supervised datasets into 80% Training and 20% Testing partitions using stratified random states.
- Standardized numerical features using standard scaling to ensure zero mean and unit variance.
- Aggregated transaction records at the User level to generate behavior profiles for clustering.

3. Regression Model: Ridge Regression (Rating Prediction)

Ridge Regression was trained to predict user product ratings. Input features included price, views, time spent, previous purchases, cart status, and product categories. Hyperparameter tuning was conducted on the regularizer strength alpha via 5-fold GridSearchCV.

Key Findings: The best regularizer strength was found at **alpha = 10.0**. The tuned model achieved an MAE of **0.5034**, RMSE of **0.5803**, and R^2 Score of **0.4508**. This indicates that user ratings are driven by page interaction duration and view frequency, allowing the business to highlight products aligned with customer taste.


Ridge Regression: Actual vs. Predicted Ratings
Perfect Prediction
Predicted Ratings
Actual Ratings
5.0
4.5
4.0
3.5
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2.0
1.5
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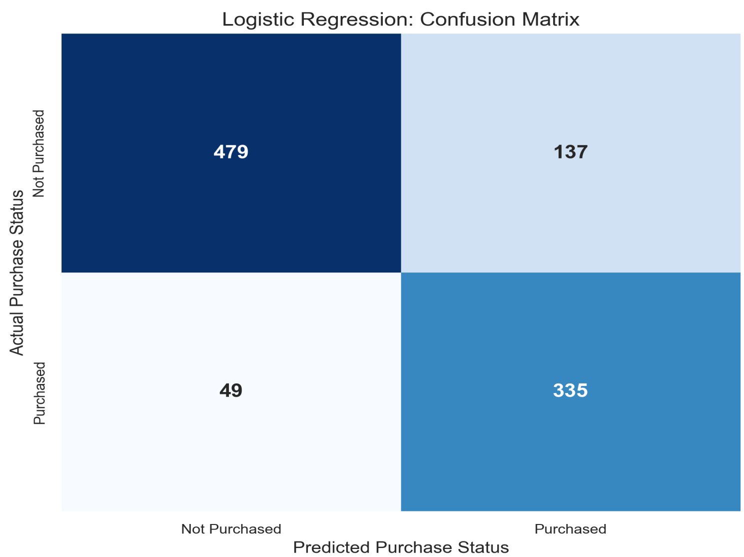
4.5
5.0

4. Classification Model: Logistic Regression (Purchase Likelihood)

Logistic Regression was implemented to forecast whether a user is likely to buy a product (Purchase_Status = 1). GridSearchCV tuned regularizer C, solver, and max_iter.

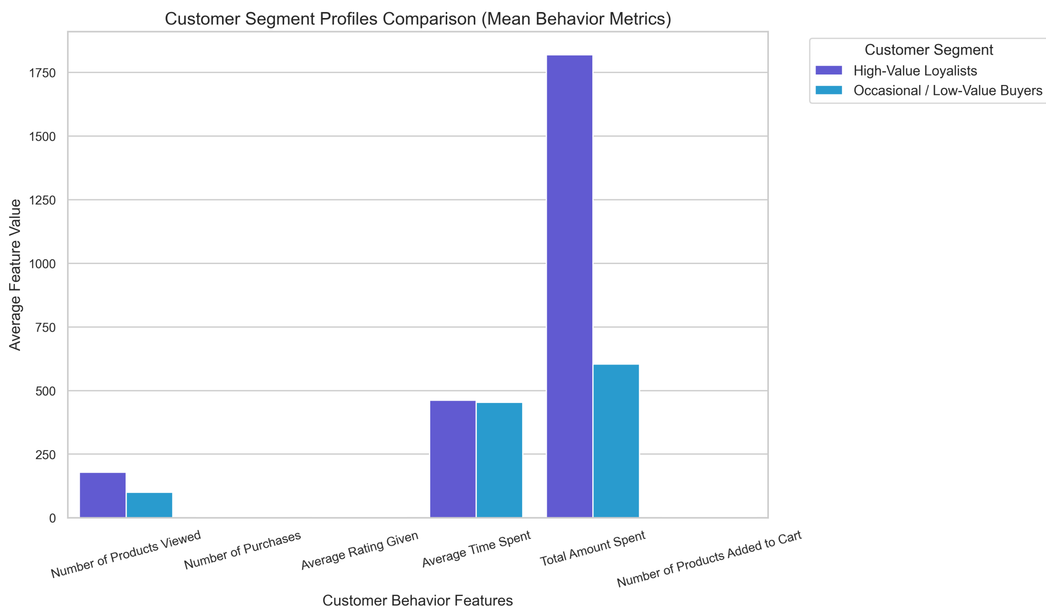
Key Findings: The best parameters identified were **C = 1**, solver = **liblinear**, and max_iter = **100**. The tuned model achieved an Accuracy of **81.40%**, Precision of **70.97%**, Recall of **87.24%**, and F1 Score of **78.27%**.



5. Clustering Model: K-Means (Customer Segmentation)

K-Means clustering grouped users into cohorts using behavioral aggregations. The Elbow Method and Silhouette scores were calculated for K values 2 through 6 to identify the optimal segment size.

Key Findings: The optimal number of segments was selected as **K = 2** based on the highest Silhouette Score of **0.2908** and the elbow bend. Each cluster corresponds to distinct shopping personas (e.g. High-Value Loyalists, Active Window Shoppers).



6. Model Comparison Table

ML Task	Algorithm	Target / Goal	Metrics Used	Best Result	Business Use Case
Regression	Ridge Regression	Predict Rating	MAE, RMSE, R^2	MAE: 0.503 RMSE: 0.580 R^2: 0.451	Recommend highly rated products to users.
Classification	Logistic Regression	Predict Purchase Status	Acc, Prec, Rec, F1	Acc: 81.4% F1: 78.3%	Target discounts to customers likely to purchase.
Clustering	K-Means	Segment Customers	Inertia, Silhouette	K: 2 Silhouette: 0.291	Tailor marketing strategies by user segment.

7. Business Insights & Recommendations

Segment-Based Recommendations:

- **High-Value Loyalists** (Size: 357): Averages 178.4 views, 3.5 purchases, and total spend of $1819.91. Action: Enroll in VIP loyalty programs, provide early-access products, and avoid aggressive discounting.
- **Occasional / Low-Value Buyers** (Size: 633): Averages 100.5 views, 1.3 purchases, and total spend of $603.55. Action: Send re-engagement offers, run seasonal clearance campaigns, and cross-sell trending items.

8. Final Conclusion

By integrating Ridge Regression (predictive ratings), Logistic Regression (purchase intent), and K-Means (demographic/behavioral grouping), the recommendation system establishes a multi-layered personalization funnel. Clustering classifies the user's cohort, Classification filters products with the highest purchase intent, and Regression ranks them to maximize engagement. Implementing this composite pipeline will optimize ad spending, elevate click-through rates, and enhance e-commerce transaction volumes.